

TELANGANA STATE-2017
CLASS 10
GENERAL SCIENCE – I

[Time: 2 hours 45 min.]

[Maximum Marks: 40]

Instructions: -

(i) Read the whole question paper and understand every question thoroughly without writing anything and 15 minutes of time is allotted for this.

Part-A

Time: 2 Hours

Marks: 35

Instructions:

(i) Answer all questions from the given **three sections I, II and III of **Part-A**.**

(ii) In Section III, every question has internal choice, answer anyone alternative.

SECTION –I

Note:-

7x1

(i) Answer all questions.

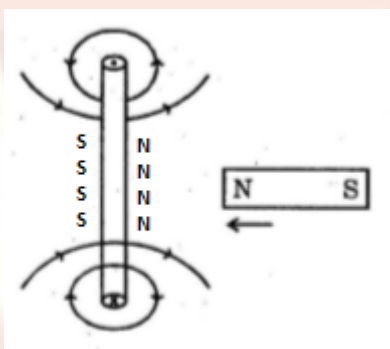
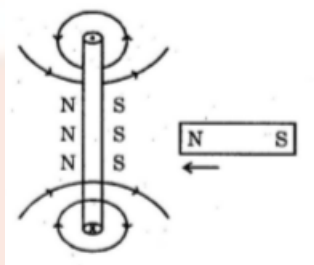
(ii) Each question carries one mark.

(iii) Write the answers in 1 to 2 sentences.

Q-1) Arrange the metals Fe, Na, Ag and Zn in increasing order of their chemical reactivity.

[Ans] $\text{Ag} < \text{Fe} < \text{Zn} < \text{Na}$.

Q-2) Correct the diagram according to Lenz law and draw it again.



Ans : this is the correct diagram

Q-3) The four quantum number values of the 21st electrons or scandium (Sc) are given in the following table.

n	l	m_l	m_s
3	2	-2	$+\frac{1}{2}$

Write the values of the four quantum numbers for the 20th electron of scandium (Sc) in form of the table.

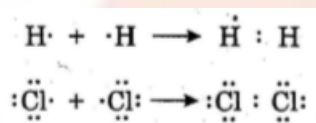
[Ans]

n	l	m_l	m_s
3	2	0	$-\frac{1}{2}$

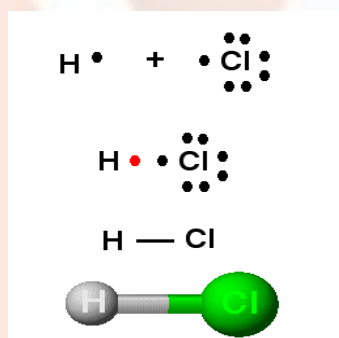
Q-4) What happens if the eye lens of a person cannot accommodate its focal length more than 2.4 cm?

[Ans] If the eye lens of a person cannot accommodate its focal length more than 2.4cm then he suffers with eye defect called Myopia.

Q-5) Show the formation of HCl molecule with Lewis dot structures using the information given below.



[Ans] The hydrogen has valency =1 and chlorine has valency =7



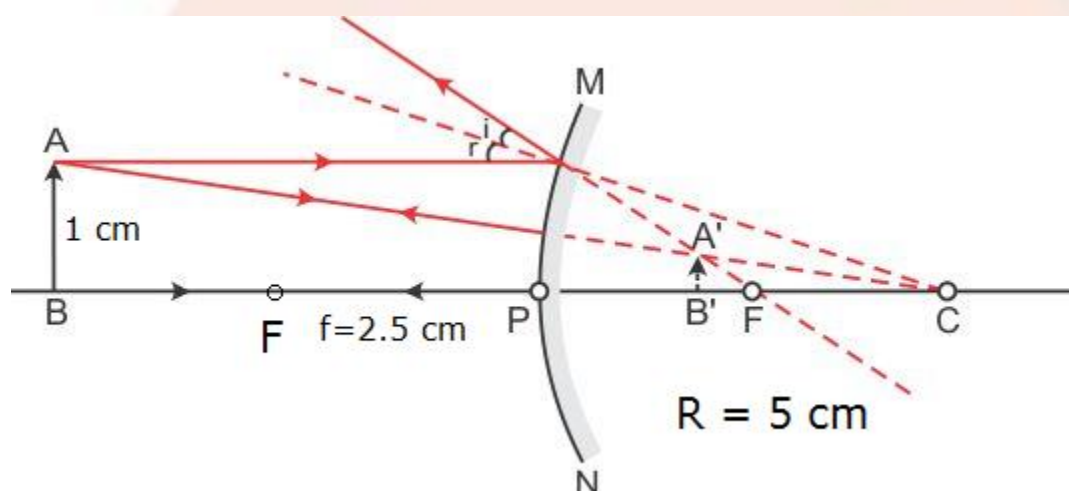
Q-6) What happens if we use a fuse made up of same wire which is used to make the electric circuit?

[Ans] If we use the same wire which is being used in electric circuit then whenever the current load increases beyond safe

levels the fuse wire will not melt and circuit will not break. This could destroy the appliances at home.

Q-7) Draw the ray diagram to show the formation of image for the object of height 1 cm. placed at 5 cm. distance, in front of a convex mirror having the radius of curvature $R = 5$ cm.

[Ans]



SECTION – II

6x2=12

Note: (i). Answer all questions.

(ii) Each question carries TWO marks.

(iii) Write the answers in 4 to 5 sentences.

Q-8) Give an example with the chemical equation for the reduction of ores using more reactive metals.

[Ans] $6\text{Na} + \text{Al}_2\text{O}_3 = 2\text{Al} + 3\text{Na}_2\text{O}$

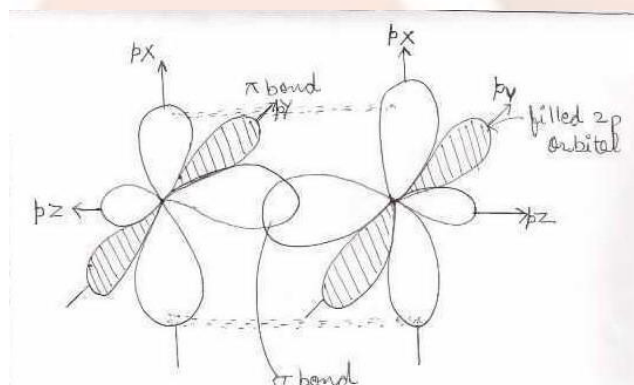
Q-9) Write the electronic configuration of the atom of an element having atomic number 11. Write the names of the rules and the laws followed by you in writing this electronic configuration.

[Ans] $\text{Na}[11] = 1s^2, 2s^2, 2p^6, 3s^1$

For electronic configuration, we follow the three important rules: Aufbau's Principle, Pauli-exclusion principle, and Hund's Rule.

Q-10) Draw the diagram to show the formation of Oxygen molecule by Valence bond theory.

[Ans]



Q-11) Observe the information given in the table and answer the questions given below the table.

Substance (in aqueous solution)	Colour change with Blue Litmus	Colour change with Red Litmus
A	Red	No change
B	No change	Blue
C	No change	No change

- (i) Which one of them may be the neutral salt among A, B, C?

[Ans] Substance C should be a Neutral Salt as it does not change the color of either blue or red litmus.

(ii) What may happen when some drops of phenolphthalein is added to the substance B?

[Ans] substance B is seen to turn Red Litmus to Blue. Therefore, it identifies as a base. As we know when phenolphthalein is added to a basic solution it turns Pink in color.

Q-12) How can you find out the focal length of concave mirror experimentally when there is no sunlight?

[Ans] By any source of light. Usually a candle does the job by using the u-v method.

Q-13) Two convex lenses of same focal length are fixed in a PVC pipe at a distance double to their focal length. What happens if a boy sees the moon with that arrangement?

[Ans] The boy will get an enlarged image of the moon that goes beyond his eyepiece.

SECTION-III

4x4

Note:

(i) Answer **all** questions.

(ii) Each question carries **FOUR** marks.

(iii) There is internal choice for each question; only one option from each question is to be attempted.

(iv) Write the answers in 8 to 10 sentences.

Q-14) Write the equation for the reaction of zinc with hydrochloric acid and balance the equation. Find out the number of molecules of hydrogen gas produced in this reaction, when 1 mole of HCl completely reacts at S.T.P

[Gram molar volume is 22.4 liters at S.T.P. ; Avogadro's number is 6.023×10^{23}]



The stoichiometric coefficients of HCl and H_2 gas are in the ratio 2:1. i.e. when 2 moles of HCl react with Zinc, only 1 mole of H_2 gas is evolved. So, when we take 1 mole of HCl we only get 0.5 moles of H_2 gas.

OR

Explain any four factors which influence the electron affinity (Electron Gain Enthalpy).

[Ans] The concept is defined as the amount of energy liberated when an electron is added to neutral gaseous atoms. Factors affecting this:

1. Nuclear Charge:

Higher the nuclear charge higher will be the attraction for the incoming electron and larger will be the value of electron Affinity.

2. Atomic Radius:

More the radius less will be the electron Affinity. Since the distance between nucleus and incoming electron increases due to increase in radius.

3. Stable electron configuration:

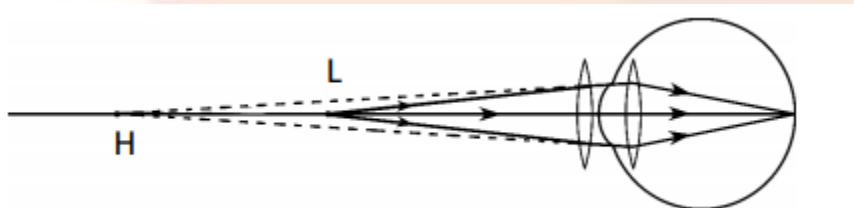
Stable the configuration of atom lesser will be it's tendency to accept an electron and hence lower the value of electron Affinity.

4. Screening Effect:

Higher the number of shells with electrons between the nucleus and its valence shell. Lesser will be the nuclear charge and lower the electron Affinity.

Q-15) How will you calculate the focal length of a biconvex lens that is used to correct the defect of Hypermetropia? Explain it mathematically.

[Ans] Hypermetropia means person with a defect to see near by objects. It is also known as long-sightedness.



Considering the above image.

Object is placed at L.

Due to the defect of Hypermetropia the image has to be formed on H by using a convex lens.

object distance = $u = -25$ cm

Image distance = $v = \text{distance of near point} = -d$

Let f be the focal length of biconvex lens

by using Lens Formula:

$$1/f = 1/v - 1/u$$

$$1/f = 1/(-d) - 1/(-25)$$

$$1/f = -1/d + 1/25$$

$$1/f = (d-25)/25d$$

Therefore

$$f = 25d/(d-25)$$

OR

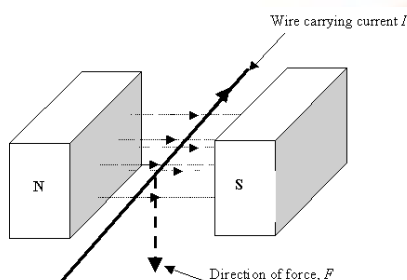
Why the current carrying straight wire which is kept in a uniform magnetic field, perpendicularly to the direction of the field bends aside? Explain this process with a diagram showing the direction of forces acting on the wire.

[Ans] Force will be proportional to the strength of the magnetic field, amount of current and the wire length when a long wire is placed in the magnetic

The magnitude of the magnetic force on a current-carrying wire is given by

$$|F_{\text{wire}}| = i|l||B|\sin\theta$$

θ = angle between the wire and the field



Q-16) List out the materials required to test whether the solutions of given acids and bases contain ions or not. Explain the procedure of the experiment.

[Ans] Aim- To find whether acid and bases contain ions or not.

Material required- Litmus paper, a solution of acid and base, gloves and eye protector.

Procedure- When you place a drop of a solution on the litmus paper, the paper changes color based on the pH of the solution. Once the color changes, you can compare it to the color chart on the paper's package to find the pH. With unknown solutions, you should wear gloves, put on eye protection and work under a fume hood to be safe. A few solutions are familiar liquids or are commonly used in science labs and it's handy to memorize what kind of solutions they are. Water and blood are both neutral.

Many household cleaners, such as bleach and ammonia, are basic, as is sodium hydroxide. Citric juices, coffee and wine are acidic. Solutions with the word "acid" in them, such as stomach acid and hydrochloric acid are acidic.

OR

List out the materials required to conduct the experiment to understand the esterification reaction. Explain the procedure of the experiment. How can you identify that an ester is formed in this reaction?

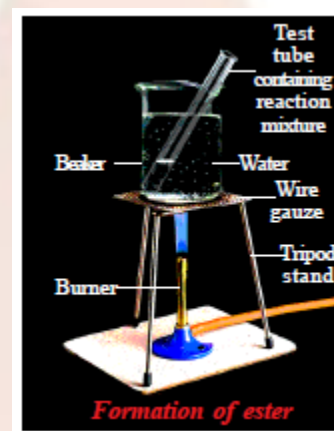
[Ans] Aim -Experiment to for the esterification reaction.

Material required- 1ml of ethanol, 1ml of glacial acetic acid, few amount of sulphuric acid, test tube, a water bag, a little amount of water and a beaker.

Procedure - Take 1 ml of ethanol (absolute alcohol) and 1ml of glacial acetic acid along with a few drops of concentrated sulphuric acid in a test tube. Warm it in a water-bath or a beaker containing water for at least five minutes as shown in Fig. Pour the warm contents into a beaker containing 20-50 ml of water and observe the odour of the resulting mixture.

Observation

You will notice that the resulting mixture is a sweet odoured



substance. This substance is nothing but ethyl acetate, an ester. The reaction in activity – 2 is called esterification reaction.

Q-17) A double concave lens with the refractive index (n) = 1.5 is kept in the air. Its two spherical surfaces have radii $R_1 = 20$ cm and $R_2 = 60$ cm. Find the focal length of the lens.

Write the characteristics of the lens.

[Ans] Given,

refractive index (n) = 1.5

radius of curvature (R_1) = -20 cm (by applying sign convention)

radius of curvature (R_2) = 60 cm (by applying sign convention)

by applying formula,

$$1/f = (n-1)(1/R_1 - 1/R_2)$$

$$1/f = (1.5-1)(1/-20 - 1/60)$$

$$1/f = (0.5)(3 + 1/-60)$$

$$1/f = (0.5)(4/-60)$$

$$1/f = (0.5)(1/-15)$$

$$1/f = 1/-30$$

$$f = -30 \text{ cm}$$

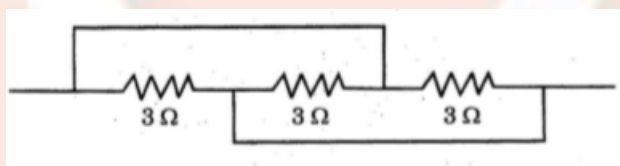
focal length of the concave lens is 30 cm.

Characteristics of the concave lens is

- 1) it will produce smaller image of object
- 2) image of the object is closer to the optical centre

OR

Find the resultant resistance for the following given arrangement. Find the current, when this arrangement is connected with 9V battery.



[Ans] Here, we know, two wires connecting in two resistances. here we use equipotential, wires have no resistance. so, potential of connecting points are same.

so, we join the connecting points in one point.

if we do it then we see

resistance combined with parallel series.

we know,

equivalent resistance (R_{eq}) of parallel series

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

then,

$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$$

$$1/R_{eq} = 3/3 = 1$$

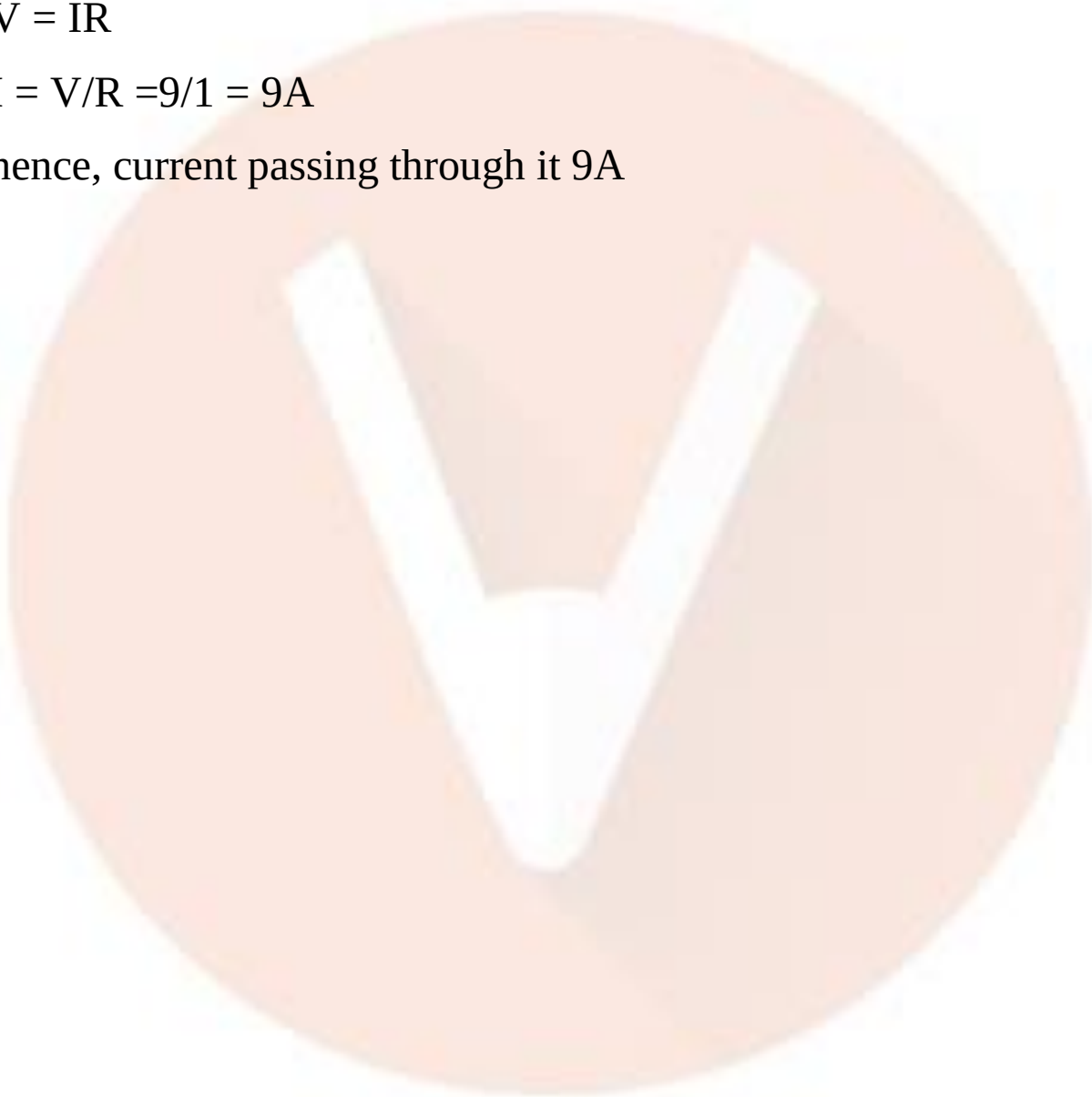
$$R_{eq} = 1 \text{ ohm}$$

use ohm's law ,

$$V = IR$$

$$I = V/R = 9/1 = 9A$$

hence, current passing through it 9A



Part-B**Time: 30 minutes****Marks: 5****Note:**

1. Answer **all** questions.
2. Each question carries $\frac{1}{2}$ marks.
3. Answers are to be written in the question paper only.
4. Marks will **not** be awarded in case of any overwriting, rewriting or erased answers.

I. Write the CAPITAL LETTER showing the correct answer for the following questions in the brackets provided against each question.

1. The reason behind the blue colour of the sky is ... [D]

- (A) Reflection of light.
- (B) Refraction of light.
- (C) Dispersion of light.
- (D) Scattering of light.

2. Ionic compound in the following is ... [C]

- (A) C_2H_6
- (B) HCl
- (C) NaCl
- (D) H_2

3. The instrument used to check the EMF of a battery is ...

[A]

- (A) Voltmeter
- (B) Ammeter
- (C) Galvanometer
- (D) Electric Tester

4. Number of moles of Oxygen needed to produce 4 moles of water on reacting with 4 moles of Hydrogen gas is [B]

- (A) 1 mole
- (B) 2 moles
- (C) 3 moles
- (D) 4 moles

5. Convex mirror is used as [C]

- (A) Shaving mirror
- (B) Focusing mirror
- (C) Rear-view mirror
- (D) Vehicle head-lights reflector

6. The electronic structure of the atom that is correctly explained by Bohr's theory in the following is [A]

- (A) Hydrogen atom
- (B) Helium atom
- (C) Carbon atom
- (D) All atoms

7. The instrument used in our daily life to convert the electrical energy into mechanical energy is [B]

- (A) Electric bulb
- (B) Electric motor
- (C) Generator
- (D) Induction stove

8. One of the uses of a Concave lens is [D]

- (A) as eye piece in microscope.
- (B) to focus the Sunlight at a point.
- (C) to correct Hypermetropia.
- (D) to correct Myopia.

9. A Dobereiner's Triad in the following is [A]

- (A) Cl, Br, I
- (B) H, He, Li
- (C) H, Na, Cl
- (D) C, N, O

10. IUPAC name of $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3$ is [D]

- (A) Chlorobutane
- (B) 2-Chlorobutane
- (C) 2, 3-Chlorobutane
- 2, 3-dichlorobutane